5.4 Electrochemistry Calculations & Applications (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.4 Electrochemistry Calculations & Applications (A Level Only)
Difficulty	Hard

Time allowed: 30

Score: /21

Percentage: /100

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Question 1a

This question is about the $Ag^+(aq) / Ag(s)$ half-cell.

A student was asked to plan an experiment to measure the standard electrode potential of the $Ag^+(aq) / Ag(s)$ half-cell.

i)
State the conditions of temperature and pressure under which standard electrode potentials are measured.

[1]

ii)
Fig. 1.1 shows the diagram drawn by the student.

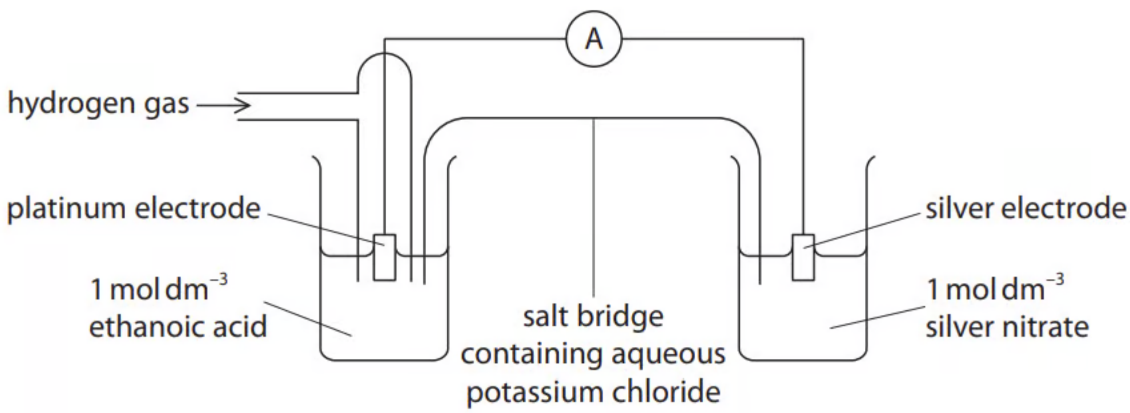

Fig. 1.1

Complete Table 1.1 to identify **three** mistakes in this diagram and the modifications that should be made to correct them.

Table 1.1

Mistake in diagram	Modification needed to correct mistake

[3]

[4 marks]

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Question 1b

The standard electrode potential, E^θ, of the Ag^+(aq) / Ag (s) half-cell is +0.80 V.

The effect of changing the concentration of the ions on the value of the electrode potential, E, in this half-cell is calculated using the equation

$$E = E^\theta + \frac{RT}{96500} \ln[Ag^+(aq)]$$

where T is the temperature in kelvin and R is the gas constant, 8.31 J K^{-1} mol^{-1}.

The electrode potential of an Ag^+(aq) / Ag (s) half-cell was measured at 20 °C and found to be +0.72 V.

Calculate the concentration of silver ions, in mol dm^{-3}, in this half-cell. Show your working.

$[Ag^+]$ = ... mol dm^{-3}

[3 marks]

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Question 2a

Table 2.1 lists electrode potentials for the $Cr_2O_7^{2-}$ (aq) / Cr^{3+}(aq) and Br_2 (l) / Br^- (aq) half-cells.

Table 2.1

Electrode reaction	***E*$^\theta$ / V**
½Br_2 (l) + e^- ⇌ Br^- (aq)	+1.09
$Cr_2O_7^{2-}$ (aq) + 14H^+ (aq) + 6e^- ⇌ 2Cr^{3+} (aq) + 7H_2O (l)	+1.36

Deduce the full equation for the $Cr_2O_7^{2-}$ (aq) / Cr^{3+}(aq) and Br_2 (l) / Br^- (aq) cell.

[1 mark]

Question 2b

Using Table 2.1, calculate the E^θ_{cell} for the electrochemical cell outlined in part **(a)**.

[1 mark]

Question 2c

The electrochemical equation for standard free energy change is given.

$$\Delta G^\theta = -nFE^\theta$$

The Faraday constant, $F = 9.65 \times 10^4$ C mol^{-1}.

Use your answer to parts **(a)** and **(b)** to determine whether the reaction of the electrochemical cell is feasible.

[1 mark]

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Question 2d

An electrochemical cell has a free energy change of -14.475 kJ mol^{-1}.

Use the information in Table 2.1 to determine the reactions taking place at each electrode of the electrochemical cell.

Table 2.1

Electrode reaction	E^{θ} / V
$Ag^+ (aq) + e^- \rightleftharpoons Ag (s)$	+0.80
$Li^+ (aq) + e^- \rightleftharpoons Li (s)$	-3.04
$ClO_2 (aq) + e^- \rightleftharpoons ClO_2^- (aq)$	+0.95
$H_2O (l) + e^- \rightleftharpoons ½H_2 (g) + OH^- (aq)$	-0.83
$Fe^{3+} (aq) + e^- \rightleftharpoons Fe^{2+} (aq)$	+0.77

- reaction at anode ....
- reaction at cathode ....

[2 marks]

Question 3a

Table 3.1 lists electrode potentials for some electrode reactions.

Table 3.1

Electrode reaction	E^{θ} / V
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+ 1.07
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+ 1.36
$[Co(H_2O)_6]^{3+} + e^- \rightleftharpoons [Co(H_2O)_6]^{2+}$	+ 1.81
$[Co(NH_3)_6]^{3+} + e^- \rightleftharpoons [Co(NH_3)_6]^{2+}$	+ 0.11
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+ 0.34
$Fe^{2+} + 2e^- \rightleftharpoons Fe$	– 0.44
$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+ 0.77
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00
$I_2 + 2e^- \rightleftharpoons 2I^-$	+ 0.54
$NO_3^- + 2H^+ + e^- \rightleftharpoons NO_2 + H_2O$	+ 0.81
$SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2 + 2H_2O$	+ 0.17
$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$	+1.00

Explain how Table 3.1 could be adapted to show an electrochemical series.

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[1 mark]

Question 3b

Use Table 3.1 to identify the halide ion that is the weakest reducing agent.

[1 mark]

Question 3c

Use Table 3.1 to justify why sulfate ions should **not** be capable of oxidising iodide ions.

[1 mark]

Question 3d

i)

Use Table 3.1 to identify an acid that will oxidise copper. Explain your answer.

[1]

ii)

Suggest a possible equation for the reaction.

[1]

iii)

Calculate the E^{θ}_{cell} for the same overall reaction.

[1]

[3 marks]

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Question 3e

Suggest why the **two** cobalt(III) complex ions in Table 3.1 have different electrode potentials.

[1 mark]

Question 3f

Use Table 3.1 to explain why $[Co(H_2O)_6]^{3+}(aq)$ will undergo a redox reaction with $[Fe(H_2O)_6]^{2+}(aq)$.

Give an equation for this reaction.

explanation ....

equation ....

[2 marks]

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Model Answers: Hard

1a

b)

i) The conditions of temperature and pressure for measuring standard electrode potentials are:

Any **one** of the following temperatures:

- 298 K
- 25 °C

Any **one** of the following pressures:

- 1 atm
- 1 bar
- 100 kPa
- 1 x 10^5 Pa
- 101 kPa
- 1.01 x 10^5 Pa
- 1 x 10^5 Nm^{-2}
- 1.01 x 10^5 Nm^{-2}

- A correct temperature
  AND
  A correct pressure; [1 mark]

ii) The mistakes and modifications are:

Mistake in diagram	Modification needed to correct mistake	
Ammeter / symbol for ammeter	Replace with (high resistance) voltmeter / symbol for voltmeter / potentiometer / Wheatstone bridge	; [1 mark]
	Replace with a solution that is 1.0 mol dm^{-3} with respect to $H^+(aq)$	

Ethanoic acid	**OR** Replace with (1.0 / 1.16–1.18 mol dm^{-3}) hydrochloric acid / HCl / nitric acid / HNO_3 **OR** Replace with 0.5 mol dm^{-3} sulfuric acid / H_2SO_4	; [1 mark]
Potassium chloride / chemical in the salt bridge	Replace with potassium nitrate / KNO_3 / sodium nitrate / $NaNO_3$ **OR** Replace the chloride anion with a nitrate anion	; [1 mark]

[Total: 4 marks]

- You are expected to know the standard conditions
  - Ion concentration of 1.00 mol dm^{-3}
  - A temperature of 298 K
  - A pressure of 100 kPa
  - Zero current
- The mistakes are:
  - The ammeter
    - This allows current to flow - it should be a high resistance voltmeter to try and achieve zero current
  - The ethanoic acid
    - All concentrations should be 1.0 mol dm^{-3} but since ethanoic acid is a weak acid it does not fully dissociate so will not achieve a concentration of 1.0 mol dm^{-3} with respect to hydrogen ions
  - The potassium chloride salt bridge
    - One of the half-cells contains silver nitrate solution which will react with the chloride ions to form a white precipitate of AgCl

1b

b) The concentration of silver ions, in mol dm^{-3}, in this half-cell is:

- Substituting values into the expression

$0.72 = 0.80 + \frac{(8.31 \times 293)}{96500} \times \ln\left[Ag^+ (aq)\right]$; [1 mark]

- Calculating ln[Ag^+ (aq)]
  ln[Ag^+ (aq)] = $(0.72 - 0.80) \times \frac{96500}{(8.31 \times 293)}$ = -3.1707; [1 mark]
- Calculating [Ag^+ (aq)]
  [Ag^+ (aq)] = $e^{\ln[Ag+(aq)]}$ = 0.041976 / 4.1976 x 10^{-2} (mol dm^{-3}); [1 mark]

[Total: 3 marks]

- When you are given equations like the one in this question, there will normally be one mark for substituting the values into the equation and one mark for rearranging the equation
- **Tip:** Take your time and make sure you put each value in the correct place
  - For questions like this, you cannot assume that a very large or very small final answer means that you have done the calculation wrong

2a

a) The full equation is:

- $Cr_2O_7^{2-}$ + 14H^+ (aq) + 6Br^- (aq) → 2Cr^{3+} (aq) + 3Br_2 (l) + 7H_2O (l); [1 mark]

[Total: 1 mark]

- The $Cr_2O_7^{2-}$ + 14H^+ (aq) + 6e^- (aq) → 2Cr^{3+} (aq) + 7H_2O (l) has the more positive E^θ value
  - So, this will be the reduction half-equation
  - The half-equation will proceed in the forward direction
- The ½Br_2 (l) + e^- → Br^- (aq) has the more negative E^θ value
  - So, this will be the oxidation half-equation
  - This means that this equation needs to be reversed
    Br^- (aq) → ½Br_2 (l) + e^-
- To combine half-equations into a full equation, the number of electrons in both equations must be equal
  - Therefore, the bromide half-equation needs to be multiplied by 6
    6Br^- (aq) → 3Br_2 (l) + 6e^-
- Now, the half-equations can be combined
  - $Cr_2O_7^{2-}$ + 14H^+ (aq) + 6Br^- (aq) + 6e^- (aq) → 2Cr^{3+} (aq) + 7H_2O (l) + 3Br_2 (l) + 6e^-
- There are 6e^- on both sides of the overall equation which can be cancelled out, to give the final overall equation
  - $Cr_2O_7^{2-}$ + 14H^+ (aq) + 6Br^- (aq) → 2Cr^{3+} (aq) + 3Br_2 (l) + 7H_2O (l)

2b

b) The value for E^{θ}_{cell} is:

$E^{\theta}_{cell} = E^{\theta}_{RHS} - E^{\theta}_{LHS}$

- E^{θ}_{cell} = 1.36 V - 1.09 V = 0.27 (V); [1 mark]

[Total: 1 mark]

- Feasible reactions have a positive E_{cell} / EMF value
- This can be calculated using $E_{cell} = E_{reduction} - E_{oxidation}$ or using $E_{cell} = = E_{right} - E_{left}$

2c

c) To determine if the reaction is feasible:

$\Delta G^{\theta} = -nFE^{\theta}$

- ΔG^{θ} = -6 mol x 96 500 C mol^{-1} x 0.27 V = -156330 J mol^{-1} / -156 kJ mol^{-1}
  AND
  ΔG^{θ} is negative and therefore spontaneous; [1 mark]

[Total: 1 mark]

- The electrochemical equation for standard free energy change may be given to you
  - The specification is unclear if this equation will definitely be given to you, so it would be worth knowing it
- **Careful:** You need 6 moles in the calculation as there are 6 electrons in the full equation
- **Remember:** ΔG^{θ} has to less than or equal to 0 for a reaction to be feasible

2d

d) To determine the reactions occurring at the anode and cathode:

$\Delta G^{\theta} = -nFE^{\theta}$

- $E^{\theta}_{cell} = \frac{\Delta G}{-nF} = \frac{-14475 \text{ J mol}^{-1}}{-1 \times 96\,500 \text{ C mol}^{-1}} = 0.15$ (V); [1 mark]
- Reaction at anode: $Ag(s) \rightleftharpoons Ag^+(aq) + e^-$
  AND
  Reaction at cathode: $ClO_2(aq) + e^- \rightleftharpoons ClO_2^-(aq)$; [1 mark]

[Total: 2 marks]

- First, you need to calculate the E^{θ}_{cell} value which can be done by rearranging $\Delta G^{\theta} = -nFE^{\ominus}$
  - $E^{\theta}_{cell} = \frac{\Delta G}{-nF}$
  - n will be equal to 1 as all of the standard electrode potentials involve the gain of 1 electron
  - Evaluation of the equation gives the value for $E^{\ominus}_{cell}$ as 0.15 V
- Working through the different combinations of electrode reactions gives one pair that have a difference / electromotive force of 0.15V
  - $Ag^+(aq) + e^- \rightleftharpoons Ag(s)$
    - This has a lower electrode potential (+0.80 V) which means that it will be the oxidation reaction occurring at the anode
    - **Careful:** The equation will need to be reversed as the question asks for the reaction that occurs at the anode
  - $ClO_2(aq) + e^- \rightleftharpoons ClO_2^-(aq)$
    - This has a higher electrode potential (+0.95 V) which means that it will be the reduction reaction occurring at the cathode

3a

a) Table 3.1 could be adapted into an electrochemical series by:

- Place / re-order the reactions into decreasing numerical order of E^{θ}
  OR
  Place / re-order the reactions into order from the highest / most positive E^{θ} value to the lowest / most negative E^{θ} value; [1 mark]

[Total: 1 mark]

- **Remember:** The electrochemical series is a list of various redox equilibria in order of decreasing E^{θ} values

3b

b) The halide ion that is the weakest reducing agent is:

- Cl^- / chloride; [1 mark]

[Total: 1 mark]

- Examiner's reports often include comments about how poorly these questions are answered
- The weakest reducing agent will be contained in the electrode reaction equation that has the highest electrode potential value
  - **Careful:** The question specifically asks for the halide ion that is the weakest reducing agent, not just the weakest reducing agent
- **Remember:** A reducing agent is itself oxidised, which means that you need to be considering the products of the electrode reaction equations
- The $Cl_2 + 2e^- \rightleftharpoons 2Cl^-$ equation has the highest electrode potential value of the halide ions
  - This means that it will proceed in the forward direction with chlorine acting as a strong oxidising agent
  - Consequently, the backward reaction is less likely to occur
  - Therefore, the chloride ion will be the weakest reducing agent

3c

of decreasing E^{θ} values

3b

b) The halide ion that is the weakest reducing agent is:

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[Total: 1 mark]

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- The $Cl_2 + 2e^- \rightleftharpoons 2Cl^-$ equation has the highest electrode potential value of the halide ions
  - This means that it will proceed in the forward direction with chlorine acting as a strong oxidising agent
  - Consequently, the backward reaction is less likely to occur
  - Therefore, the chloride ion will be the weakest reducing agent

3c

c) Sulfate ions should **not** be capable of oxidising iodide ions because:

- The electrode potential for sulfate ions / sulfur dioxide is less than that for iodine / iodide
  OR
  E^{θ} SO_4^{2-} / SO_2 < E^{θ} I_2 / I^-; [1 mark]

[Total: 1 mark]

- The relevant half-equations for this question are:
  - $I_2 + 2e^- \rightleftharpoons 2I^-$ with an E^{θ} value of + 0.54 V
  - $SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2 + 2H_2O$ with an E^{θ} value of + 0.17 V
- From these half-equations, the iodine / I_2 half-equation should proceed in the

forward direction as it has the higher E^{θ} value
  - While the sulfate ion / SO_4^{2-} half-equation should proceed in the backward direction as it has the lower E^{θ} value
- This means that iodine / I_2 will be reduced to iodide ions / I^- while sulfur dioxide / SO_2 is oxidised to sulfate ions / SO_4^{2-}

3d

d)

i) The acid that will oxidise copper is:

- Nitric acid / HNO_3 **AND** because the electrode potential for nitrate ions / NO_3^- is more than that for copper(II) ions / Cu^{2+}
  OR
  Nitric acid / HNO_3 **AND** because NO_3^- has a more positive E^{θ} than Cu^{2+}
  OR
  Nitric acid / HNO_3 **AND** because E^{θ} NO_3^- (/NO) > E^{θ} Cu^{2+}(/Cu)
  OR
  Nitric acid / HNO_3 **AND** because NO_3^- is a better oxidising agent than Cu^{2+}; [1 mark]

ii) The equation for this reaction is:

- $Cu + 2NO_3^- + 4H^+ \rightleftharpoons Cu^{2+} + 2NO_2 + 2H_2O$
  OR
  $Cu + 2HNO_3 + 2H^+ \rightleftharpoons Cu^{2+} + 2NO_2 + 2H_2O$; [1 mark]

iii) E^{θ}_{cell} for this reaction is:

- + 0.47 (V); [1 mark]

[Total: 4 marks]

- The possible acids that could be formed from the various ions in the table are:
  - Hydrobromic acid / HBr
  - Hydrochloric acid / HCl
  - Nitric acid / HNO_3
  - Sulfuric acid / H_2SO_4
- The oxidation of copper requires the $Cu^{2+} + 2e^- \rightleftharpoons Cu$ half-equation, with an E^{θ}

value of + 0.34 V, to be driven backwards
  - The half-equation for the hydrogen ion has a lower E^θ value
  - This means that the copper(II) ions / Cu^{2+} would be reduced to copper / Cu while hydrogen / H_2 is oxidised to hydrogen ions / H^+
  - Therefore, it must be the other ion from the acid that can cause the oxidation of copper / Cu
- The E^θ value of the sulfate ion / $SO_4{}^{2-}$ half-equation is lower than the copper(II) / Cu^{2+} half-equation
  - This means that the copper(II) ions / Cu^{2+} would be reduced to copper / Cu while sulfur dioxide is oxidised to sulfate ions / $SO_4{}^{2-}$
  - Therefore, sulfuric acid / H_2SO_4 cannot cause the oxidation of copper / Cu
- The E^θ value of the bromine / Br_2 and chlorine / Cl_2 half-equations are both higher than the copper(II) / Cu^{2+} half-equation
  - This means that the bromine / Br_2 and chlorine / Cl_2 would be reduced to their halide ions, instead of the halide ions from the acids reacting
  - These reductions mean that hydrobromic acid / HBr and hydrochloric acid / HCl cannot cause the oxidation of copper / Cu
- The E^θ value of the nitrate ion / $NO_3{}^-$ half-equation is higher than the copper(II) / Cu^{2+} half-equation
  - This means that the nitrate ion / $NO_3{}^-$ would be reduced to nitrogen dioxide / NO_2
  - This would cause the oxidation of copper / Cu to copper(II) ions / Cu^{2+}
- This means that the relevant half-equations for parts (ii) and (iii) are:
  - $Cu^{2+} + 2e^- \rightleftharpoons Cu$
  - $NO_3{}^- + 2H^+ + e^- \rightleftharpoons NO_2 + H_2O$
- To combine the half-equations:
  - The copper(II) ion / Cu^{2+} half-equation needs to be reversed as it has the lower E^θ value
    - $Cu \rightleftharpoons Cu^{2+} + 2e^-$
  - The nitrate ion / $NO_3{}^-$ half-equation needs to be doubled so that both half-equations have the same number of electrons
    - $2NO_3{}^- + 4H^+ + 2e^- \rightleftharpoons 2NO_2 + 2H_2O$
  - The half-equations can now be combined
    - $Cu + 2NO_3{}^- + 4H^+ + 2e^- \rightleftharpoons Cu^{2+} + 2NO_2 + 2H_2O + 2e^-$
  - Cancelling the electrons on both sides gives the overall equation
    - $Cu + 2NO_3{}^- + 4H^+ \rightleftharpoons Cu^{2+} + 2NO_2 + 2H_2O$
- To calculate E^θ_{cell}:
  - **Remember:** $E^\theta_{cell} = E^\theta_{reduction} - E^\theta_{oxidation}$

- The nitrate ion / NO_3^- half-equation is the reduction half-equation, while the copper(II) ion / Cu^{2+} half-equation is the oxidation half-equation
- Therefore, E^θ_{cell} = (+ 0.81) – (+ 0.34) = + 0.47 V

3e

e) The **two** cobalt(III) complex ions in Table 3.1 have different electrode potentials because:

- They have different ligands; [1 mark]

[Total: 1 mark]

- Ideas about different concentrations or oxidation states would be ignored / rejected for this answer

3f

f) $[Co(H_2O)_6]^{3+}$(aq) will undergo a redox reaction with $[Fe(H_2O)_6]^{2+}$(aq) because:

- The electrode potential for the cobalt(III) / Co^{3+} ions is more / greater than that for the iron(III) / Fe^{3+} ions
  OR
  E^θ Co^{3+} > Fe^{3+}
  OR
  1.81 > 0.77
  OR
- E^θ_{cell} = + 1.04 V; [1 mark]

The equation for the reaction is:

- $[Co(H_2O)_6]^{3+} + [Fe(H_2O)_6]^{2+} \rightleftharpoons [Co(H_2O)_6]^{2+} + [Fe(H_2O)_6]^{3+}$
  OR
  $Co^{3+} + Fe^{2+} \rightleftharpoons Co^{2+} + Fe^{3+}$; [1 mark]

[Total: 2 marks]

- This question is another application of how the electrode potential value of a half-equation relates to its relative reactivity
- There is an unusual twist in this question with the inclusion of the cobalt(III)

complexes in the table, but not all of the transition metals have their equations written with complexes
- Therefore, you are required to recognise that the $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ is another way of writing $[Fe(H_2O)_6]^{3+} + e^- \rightleftharpoons [Fe(H_2O)_6]^{2+}$

- The electrode reactions that are relevant to this question are:
  - $[Co(H_2O)_6]^{3+} + e^- \rightleftharpoons [Co(H_2O)_6]^{2+}$ with an E^θ value of + 1.81 V
  - $Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$ with an E^θ value of + 0.77 V
- From these half-equations, the cobalt(III) half-equation should proceed in the forward direction as it has the higher E^θ value
  - While the iron(III) half-equation should proceed in the backward direction as it has the lower E^θ value
- This means that cobalt(III) / $[Co(H_2O)_6]^{3+}$ / Co^{3+} will be reduced to cobalt(II) / $[Co(H_2O)_6]^{2+}$ / Co^{2+} while iron(II) / $[Fe(H_2O)_6]^{2+}$ / Fe^{2+} is oxidised to iron(III) / $[Fe(H_2O)_6]^{3+}$ / Fe^{3+}
  - $[Co(H_2O)_6]^{3+} + e^- \rightleftharpoons [Co(H_2O)_6]^{2+}$
  - $[Fe(H_2O)_6]^{2+} \rightleftharpoons [Fe(H_2O)_6]^{3+} + e^-$
- Combining these half-equations and cancelling the electrons gives the overall reaction equation
  - $[Co(H_2O)_6]^{3+} + [Fe(H_2O)_6]^{2+} \rightleftharpoons [Co(H_2O)_6]^{2+} + [Fe(H_2O)_6]^{3+}$
    This can also be written without the ligands as $Co^{3+} + Fe^{2+} \rightleftharpoons Co^{2+} + Fe^{3+}$